

Fractus — User Manual

Granular Processor by The WAV Lab Version 0.9 public beta — manual revision 1.6, 28 September 2026

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1. Welcome

Fractus is a granular processor. It continuously records the audio passing through it into a buffer. It then plays that recording back as a cloud of short overlapping fragments called **grains**. Each grain has its own position in the recording, pitch, direction, stereo placement and envelope. Change how grains are made and the result can be a subtle shimmer, a frozen pad, a stuttering rhythm or a dense wash of texture.

After the grains, the signal passes through a **Character** stage (tape and converter-style colour), a **diffuser** that smooths grain edges, and a **reverb**. An 8-slot **modulation matrix** can move almost any control. Optional **MIDI modes** turn Fractus into a playable, pitch-tracking instrument without it ceasing to be an effect.

The name comes from *fractus*, a species of ragged, broken cloud, from the Latin for "broken". Breaking sound apart is what Fractus does.

What sets Fractus apart

- **Up to 256 grains at once.** The hardware granular modules that inspired it were limited by small processors; Fractus is not.
 - **A 30-second buffer** you can freeze, save to a file, or load from one.
 - **Per-grain stereo placement, reverse playback and pitch.**
 - **Tempo sync** of grain triggers, grain length, buffer length and modulation, phase-locked to your host's timeline.
 - **Three clean-room reverbs**, every parameter of which can be modulated.
 - **Level-honest design.** Density, grain size, the Character macro and reverb decay do not double as volume controls. The default patch sits within 1 dB of its own input, so bypassing it is not a level jump.
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2. Installation and system requirements

Formats

Format	macOS	Windows
VST3	Yes	Yes
AAX (Pro Tools)	Yes	Yes
Audio Unit (AU)	Yes	—

Requirements

- **macOS:** macOS 12 Monterey or later, **Apple Silicon (M-series) only**. Intel Macs are not supported.
- **Windows:** Windows 10 or later, 64-bit. The installer will not run on older versions.
- A host that supports VST3, AAX or AU.

Installing

Download the installer for your computer from the Fractus page at thewavlab.com/product/fractus/. The same installer is used for the trial and the full version; buying Fractus unlocks the copy you already have.

macOS (the .pkg installer)

1. **Quit your DAW.**
2. Double-click the installer. It is signed and notarized by Apple, so it opens without a security warning.
3. Accept the licence agreement. All three formats (VST3, AAX, Audio Unit) are installed by default; to choose, click **Customize** on the Installation Type step.
4. Enter your Mac password when asked. The plug-ins go into the system plug-in folders, so every user account on the Mac can use them.

Windows (the Setup.exe installer)

1. **Quit your DAW.** If Fractus is still loaded in a program, Setup stops and asks you to close it, because Windows cannot replace a plug-in that is in use.
2. Run the installer and allow it to make changes when Windows asks. It is signed by Sounds Like Joe LLC.
3. Accept the licence agreement, then choose **Full** (VST3 and AAX), **VST3 only** or **AAX only**.

Installing again, or installing a newer version, replaces the old copy completely. To remove Fractus on Windows, use **Settings > Apps**; on macOS, delete the files from the locations below.

Fractus installs to the standard locations:

- **VST3:** /Library/Audio/Plug-Ins/VST3 (macOS), C:\Program Files\Common Files\VST3 (Windows)
- **AAX:** /Library/Application Support/Avid/Audio/Plug-Ins (macOS), C:\Program Files\Common Files\Avid\Audio\Plug-Ins (Windows)
- **AU:** /Library/Audio/Plug-Ins/Components

After installing or updating, fully quit and restart your DAW. Most hosts keep the old version loaded until they restart, even if you remove and re-insert the plugin.

Where to find Fractus in your host

Fractus is an **audio effect**. It appears in your effect or insert list, not among instruments.

- It works on **stereo tracks**, and on **mono tracks** as a mono-in, stereo-out insert.
- In **Logic Pro** it is listed as a "MIDI-controlled effect" (music effect) so it can receive MIDI. You will find it in the Audio FX slots under **Audio Units > The WAV Lab**.
- In **Pro Tools** it appears under the WAV Lab category for both mono and stereo tracks.

GarageBand: Fractus loads directly, with no security prompt, and shares its presets and licence with your other music software.

3. Free trial, activation and demo mode

The two pages you'll use are **thewavlab.com/fractus/trial/** (the free trial) and **thewavlab.com/fractus/unlock/** (unlocking a purchase and resetting your computers). The buttons in the licence panel open them for you.

Fractus never connects to the internet. Unlocking it is a copy-and-paste between the plugin and a page on our website, so it also works on a studio computer that is never online: take the machine code to any computer with a web browser.

Your machine code

Click **LICENCE** in the top-right corner of the Fractus window. The panel shows this computer's **machine code**, for example **FRCT-7QK2-M9XD-4HTA-P3RN**, with a **Copy** button. Each computer has its own code. The code never uses the letters I, L, O or U, so it can't be misread.

The word in the corner tells you the licence state at a glance: **DEMO** before a trial or after it ends, **TRIAL 9D** during a trial (the days left), and **LICENCE** once Fractus is bought.

14-day free trial

1. In the licence panel, click **Start 14-Day Trial**. The trial page opens in your web browser.
2. Paste your machine code and click **Get Trial Code**. You don't need an account or an email address.
3. Copy the response code the page shows, paste it into the licence panel and click **Unlock**.
4. Fractus runs fully, with no interruptions, for 14 days. The licence panel shows how many days are left.

Each computer can have one trial. Reinstalling Fractus doesn't start a new one.

Demo mode

Before a trial starts, and after it ends, Fractus runs in **demo mode**. It is fully functional, but every 30 seconds the output fades to silence for a fraction of a second. The first dropout comes 30 seconds after the plugin starts.

Activating a licence

When you buy Fractus, your **licence key** is emailed to you.

1. Open the licence panel and copy your machine code.
2. Go to the **unlock page**, thewavlab.com/fractus/unlock/ (or click **Open Unlock Page** in the licence panel). Enter your licence key and machine code, then click **Unlock**.
3. Copy the response code, paste it into the licence panel and click **Unlock**. You can do this during or after your trial.

Fractus is now unlocked on that computer for good. The licence panel and the About box show the licence holder's name.

A licence covers two computers. To move Fractus to a new computer, or after reinstalling your operating system (which can change the machine code), use **Reset my computers** on the unlock page. You can do this once a month, then unlock the new machine. If you need more resets, contact support.

Lost the unlock? Enter your key and the same machine code on the unlock page again. You'll get the same response code, and it doesn't use up one of your two computers.

Instances that were already open when you unlocked stay in demo mode until you reload them or reopen the session; the licence panel reminds you after a successful unlock. New instances are unlocked straight away.

If a code is refused, the panel says why: a code for a different computer, an expired trial code, a computer clock set to the wrong date, or a code that was only partly copied. Copy the whole code, which starts with **FRACTUS1-**; line breaks added by an email program don't matter.

4. Quick start

1. **Insert Fractus** on a track with some sustained or melodic material: a pad, a vocal, a guitar part.
2. **Play the track.** Out of the box you will hear a gentle granular shimmer with a touch of reverb. The waveform in the display shows what Fractus has recorded, and the dots moving across it are the grains.
3. **Raise Jitter.** Grains now pick positions scattered across the buffer, and the sound turns into a cloud.
4. **Move Position.** It sets how far back in time grains read. The highlighted band in the display follows it.

5. **Change Pitch.** Try +12 or -12 semitones.
6. **Change Grain Size and Density.** Short grains at high density make fizz and texture. Long grains at low density make overlapping echoes.
7. **Press Freeze.** Recording stops and the display goes still. The grains now play from what is held in the buffer, however the input changes.
8. **Turn up Character** to add grit and lo-fi colour.

To get back to the start, reload the plugin or load the default preset.

5. How Fractus works

The signal path

input → BUFFER → GRAIN ENGINE → (Grain Mix) → CHARACTER → DIFFUSER → REVERB → OUTPUT TRIM → output

1. **Buffer.** Your audio is recorded continuously into a stereo buffer up to 30 seconds long. Recording pauses while Freeze is on.
2. **Grain engine.** Grains are started ("triggered") at a rate you choose. Each one plays a short piece of the buffer through an envelope, with its own pitch, direction and pan. Fractus balances the level as grains overlap, so the overall loudness stays steady.
3. **Grain Mix** blends your original, unprocessed input back in at this point.
4. **Character** adds saturation, band-limiting, sample-rate reduction and bit-crushing. It colours what is read *out* of the buffer and never touches what is stored.
5. **Diffuser** smooths and blurs grain edges.
6. **Reverb:** Plate, FDN or Early Reflections.
7. **Output Trim** is a final gain adjustment.

Four ideas that explain most of Fractus

- **Position is a place in time, measured back from "now".** 0% is the newest audio and 100% is the oldest the buffer holds. Position and Jitter are percentages of the buffer, so if you change Buffer Length, the same Position points at a different moment.
- **Overlap is what you hear.** Overlap is how many grains sound at once, roughly *Grain Size × Density*. At low overlap you hear individual grains. At high overlap they fuse into a continuous texture.
- **Grains are set when they start.** Position, Jitter, Pitch, Pan, Scatter, Grain Shape and Reverse are fixed for each grain at the moment it starts. Moving those controls, by hand or by modulation, changes the next grains and never clicks the ones already playing.

- **Everything stays in the recorded region.** A grain never reads past the newest audio or wraps round to the oldest. If there is not room for a grain at its full length, it is shortened with a proper fade rather than allowed to click.
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6. The interface

Fractus has **three pages**, chosen with the tabs at the top of the window:

- **MAIN:** the sound. Buffer, triggering, grains, space, Character, diffuser and reverb, plus the display.
- **MODULATION:** the modulation matrix and its sources (four LFOs, an envelope follower, two random generators).
- **ADVANCED:** Character's five stages, MIDI, the buffer's save and load functions, and presets.

The **top-right corner** holds **LICENCE** (machine code, trial and unlock), the **day/night theme** toggle and **About**.

Working with controls

- **Drag** a knob or slider to change it.
- **Double-click the value** under a control (the number, not the knob) to type a value, then press Return. Type in the units shown: "35" on a percentage control means 35%. Decimals are accepted and rounded to the nearest legal value. Text that isn't a number is ignored and the control keeps its value.
- **Right-click** any control for **MIDI Learn** (see Chapter 10).
- **Every control can be automated** in your DAW, including typed values, which record like any other change.

Ghosted controls

Some controls appear **dimmed** at times. A dimmed control is *not* disabled: it still works and can still be automated. It is dimmed because another setting is currently cancelling its effect, and the thing to change is usually close by. Examples:

- **Grain Size** dims when Grain Size Sync is on, because the Division sets the length instead.
- **Density** dims in Sync trigger mode, where the Trigger Division sets the rate. It also dims when Max Grains is limiting the rate (see Chapter 13).
- **Character's five stages** dim while the Character macro is at zero. The macro dims if all five stages are at zero.

- **The MIDI controls** dim while MIDI Mode is Off.
- A **matrix slot's** Destination and Depth dim while its Source is None.

Window size and theme

Drag the window's corner to resize. Everything scales together. The **sun/moon** icon switches between the **night** and **day** themes. Your theme choice is saved with the session, and loading a preset never changes it.

7. MAIN page

The MAIN page reads left to right in signal order: **Buffer and Trigger** | **Grain** | **Space and Character** | **Diffuser and Reverb**, with the display across the top.

7.1 The display

The display shows what the grains are reading.

- **The waveform** is the contents of the buffer, **before any processing**: Character, the diffuser and the reverb never appear in it. The label *BUFFER · PRE-CHARACTER* is a reminder of this. **The newest audio is at the left** and older audio runs to the right, matching the Position strip below.
- **The dimmed region** is the part of the buffer that has not been recorded yet. It appears after inserting the plugin and after loading or reloading.
- **The highlighted band** is Position $\pm$ Jitter, the region grains are chosen from.
- **Grains** are the points of light. Horizontal position is where each grain is reading. Vertical position is its **pan**, from left at the top to right at the bottom. Brightness follows each grain's envelope. **Each grain trails a faint tail** back to where it started: forward grains trail one way and reverse grains the other.
- **When Freeze is on**, the waveform stops moving.
- **At very short buffer lengths** (below about a tenth of a second), the waveform fades out and only the grains remain.
- **The output meter** runs down the display's right edge, next to the **Output Trim** slider.

7.2 Position strip

The slider directly under the display is **Position**. It shares the display's horizontal axis, so dragging it slides the highlighted band.

7.3 BUFFER

Control	What it does
Length	How much recent audio the buffer holds, from 0.05 s to 30 s. Default 4 s.
Sync	Sets the length in bars from your host's tempo and time signature instead of in seconds.
Bars	The synced length: 1/4 bar to 16 bars. Default 2 bars (4 seconds at 120 bpm in 4/4).
Freeze	Stops recording. Grains play from whatever the buffer holds, whatever the input does.

About the buffer

- It starts **empty** when the plugin loads and fills over one buffer length. During that time grains read only the audio that exists, so there is sound straight away.
- **Freeze** is the headline feature. Freeze a chord, a word or a hit, then shape it with every other control.
- A frozen buffer is **saved with your session** (Chapter 11).
- Changing the **audio buffer size** in your DAW does not lose recorded or frozen audio. Changing the **sample rate** empties the buffer (see Chapter 13).
- Buffer Length is deliberately **not** a modulation destination, because resizing a buffer while it is being read would glitch.

7.4 TRIGGER

Control	What it does
Mode	Internal: a steady clock at the Density rate. Sync: grains land on your host's note grid. Random: grains arrive at random times whose average rate is set by Density.
Division	The note value used in Sync mode, from 1/1 to 1/64, including dotted and triplet values. Default 1/16.

- **Sync mode locks to the beat.** Grains land on the actual grid positions, on the same beats each time a loop repeats, and after you jump to a new position.
- **With the transport stopped,** Sync mode keeps running at the tempo-derived rate, so you can audition while stopped.
- **Random mode** clusters and thins naturally, as a real cloud does. It is a true random process, not a regular clock with wobble added.
- The Division list is ordered from longest to shortest. Turning the control one way always makes the interval shorter.

7.5 GRAIN

Control	What it does
Size	Grain length, from 1 ms to 4000 ms. Default 80 ms. The most useful range is roughly 1–200 ms.
Size Sync / Division	Sets grain length as a note value instead of milliseconds.
Density	How many grains start per second in Internal and Random modes, from 0.1 to 1000 Hz. Default 40 Hz.
Pitch	Transposition in semitones, from –48 to +48. Grain <i>length</i> is unchanged, so this is pitch-shifting without time-stretching.
Reverse	The chance that each grain plays backwards, from 0% (all forward) to 100% (all reversed).
Shape	Grain envelope. 0% is a square (boxcar) window, 50% a triangle and 100% a smooth Hann bell. The small picture above the control shows the current shape.
Max Grains	The most grains that can sound at once, from 1 to 256. Default 32.
Grain Mix	Blend between your dry input (0%) and the grains (100%).

Size. The long end of the range is there for tempo-synced grains: a synced whole note is 4 seconds at 60 bpm. Grains longer than the trigger interval overlap into a continuous sound, while grains shorter than it leave gaps, which is how you get rhythmic chopping.

Reverse plays the **same stretch of audio backwards**, like backward tape. It does not read into older audio. Put Position on a drum hit and reverse grains swell *into* the hit. Settings in between give a mixed cloud with the occasional backward grain.

Max Grains is a CPU budget. Past roughly 30 to 60 overlapping grains the texture stops changing and only the CPU meter moves, so set it for the character you want rather than turning it up on principle. A full 16-note chord with half-second grains reaches 256.

Max Grains limits Density. When $\text{Density} \times \text{Size}$ would need more grains than Max Grains allows, grains keep their full length and simply arrive less often. Density dims, and its readout shows the rate actually being delivered. In Sync mode the grid thins evenly instead (every 2nd, 3rd or 4th point) and the Division dims. Raise Max Grains to get the full density back.

Grain Mix brings the dry signal in *before* Character, so Character colours the dry too. In the gated MIDI modes, the dry is silenced along with the grains when no note is held.

7.6 SPACE

Control	What it does
Jitter	Random variation around Position, as a percentage of the buffer.
Scatter	Random stereo placement of each grain around Pan. At 0% every grain sits at Pan. At 100% grains spread across the whole stereo field.
Pan	Where the grain cloud is centred in the stereo field.

The two pairs mirror each other:

- **Position** is where grains read in the buffer, and **Jitter** randomises it.
- **Pan** is where grains sit in the stereo field, and **Scatter** randomises it.

Pan is fixed for each grain when it starts, so an LFO on Pan does not sweep the output. It lays grains down at moving positions, leaving a *trail* across the stereo field, which a track's own pan control cannot do.

7.7 CHARACTER

Control	What it does
Amount	The Character macro, from 0% (clean) to 100% (maximum lo-fi).
STAGES	Jumps to the ADVANCED page, where you set how far each of the five stages goes.

Character is a **settings control, not a wet/dry blend**. Turning it up pushes every stage further: more drive, narrower bandwidth, lower sample rate and fewer bits, the way a cheaper converter degrades sound. At 0% the stage is completely transparent. It is **loudness-compensated**, so turning it up changes the tone, not the volume. Character never alters what is stored in the buffer: turn it back down and the clean material is still there.

7.8 DIFFUSER

Control	What it does
Mix	How much diffusion is applied. Default 35%.
Feedback	Feeds the diffuser's output back into itself for longer, smeared tails.
Crossfeed	Blends left and right inside the diffuser, merging the two sides.

The diffuser softens grain edges and blurs the cloud without changing its tone. Its feedback stays inside the diffuser and reverb and never reaches the recording buffer.

7.9 REVERB

Control	What it does
Type	Plate: dense and smooth. FDN: a spacious room or hall. Early Reflections: a short space with no long tail.
Decay	Tail length.
Pre-Delay	Gap before the reverb begins, from 0 to 250 ms.

Control	What it does
Damping	Darkens the tail.
Size	Size of the modelled space.
Mix	How much reverb you hear. Default 25%.

Decay controls length, not loudness: a long tail is no louder than a short one. Damping darkens the tail and so makes it a little quieter. Every reverb control, including Type, can be modulated. Character's Reverb Tone stage also darkens the reverb as Character rises.

7.10 Output Trim

A vertical slider inside the display, from -12 to $+12$ dB. Use it on hot or heavily compressed sources: a grain cloud can peak a few dB above the original even at the same average level.

8. MODULATION page

8.1 The matrix

Eight slots, each with a **Source**, a **Destination** and a **Depth**:

- **Source:** None, LFO 1–4, Follower, Random 1–2 or Amp Envelope.
- **Destination:** 26 controls, including Grain Size, Density, Pitch, Position, Jitter, Pan, Scatter, Grain Shape, Max Grains, Reverse, Grain Mix, every diffuser and reverb control, Character and its five stages, and Output Trim.
- **Depth:** -100% to $+100\%$. A negative depth inverts the source.

How modulation behaves

- **Modulation is added on top of the control's own setting.** The knob keeps your setting and the modulation moves the sound around it. Because the knob itself is never moved, **modulation never fights your DAW automation:** automation sets the base and modulation moves it.
- **A depth of 100% sweeps the destination's full range.** Modulation follows the control's own curve, so modulating Grain Size behaves the way the knob does under your hand.
- **Slots add together.** Two slots aimed at the same destination combine, and equal but opposite depths cancel.
- **One source, several destinations.** Aim one LFO at two controls with opposite depths and they move against each other.

- Each row shows the destination's **live value**, and a coloured **ring** appears on the destination control on every page. Colour identifies the source: blues for the LFOs, pinks for the randoms, a warm colour for the Follower and a pale colour for the Amp Envelope. The source's name always appears alongside.

Modulation updates once per audio block. At common buffer sizes that is about 90 updates a second, which is plenty for grain parameters. They are set when each grain starts anyway. At very large host buffers (2048 samples and above), fast LFO rates above about 20 Hz lose accuracy.

8.2 LFO 1–4

Control	What it does
Rate	0.01 to 20 Hz.
Shape	Sine, Triangle, Saw Up, Saw Down or Square. The icon shows the shape.
Sync	Locks the LFO to host tempo.
Division	The synced cycle length, as a note value.

Synced LFOs lock to your host's timeline and restart in the same place each time a loop repeats. With the transport stopped they keep running at the tempo-derived rate.

8.3 Follower (envelope follower)

The Follower tracks the level of Fractus's **input** and turns it into a modulation signal from 0 to 1.

Control	What it does
Attack	How quickly the Follower rises, from 0.1 ms to 2 s.
Release	How quickly it falls, from 0.1 ms to 2 s.
Gain	Boosts quiet sources so they reach the full range, from 0.1× to 16×.

The Follower always listens to the input, never to Fractus's own output. That prevents runaway feedback loops.

Try: Follower → Density, so louder playing makes a denser cloud. Or Follower → Reverb Mix at negative depth, which "ducks" the reverb while you play.

8.4 Random 1 and 2

Sample-and-hold random generators.

Control	What it does
Rate	How often a new random value is chosen, from 0.01 to 50 Hz.
Sync / Division	Locks the rate to host tempo.
Slew	0% is stepped: each value jumps. Higher settings glide smoothly between values.

Synced random generators repeat each time a loop repeats, so a random pattern becomes part of your patch. Unsynced ones never repeat.

8.5 Amp Envelope as a source

In the MIDI modes, the note's amplitude envelope (Chapter 10) can also drive any destination. Try it on Reverb Mix or Character, so each note blooms as it plays.

9. ADVANCED page

9.1 Character stages

The **Character** macro is repeated here, followed by five stage controls, each from 0% to 100%:

Stage	Effect at full
Saturation	Soft drive.
Band Limit	A narrowing band, about 300 Hz to 2 kHz at full: a cheap converter's input, close to a telephone.
Decimate	Lowers the sample rate, adding aliasing grit.

Stage	Effect at full
Bit Crush	Reduces bit depth, adding quantisation crunch.
Reverb Tone	How much Character darkens the reverb.

How it works: each stage's effect = **Character × its stage control**. The stage controls set the ceiling for each effect, and the macro sweeps all five from zero up to their ceilings together. All five default to 100%, so Character alone uses everything. Set a stage to 0% to leave it out of the macro. Each stage's range is tapered to where that effect is actually audible.

Interactions worth knowing:

- **More Saturation makes Bit Crush sound *cleaner*.** Driving the signal harder uses more of the available bit levels. For maximum crunch, use low Saturation with high Bit Crush. For driven but clearer, raise both.
- **Band Limit changes how gritty Decimate sounds,** because filtering before sample-rate reduction limits aliasing, as in a real converter.
- **With Character at 0%, the stage controls do nothing.** They dim to show this.

9.2 BUFFER (sample manager)

Control	What it does
Freeze	The same Freeze as on MAIN, repeated here.
Save Buffer	Writes the frozen buffer to a 32-bit float WAV file. Available only while frozen.
Load Buffer	Loads an audio file into the buffer and turns Freeze on.

- **Save needs Freeze,** so that what you save is exactly what you are hearing.
- **Load turns Freeze on for you.** It accepts WAV, AIFF, FLAC and MP3 at any bit depth and sample rate, converting as needed.
- **A file longer than the buffer:** if Buffer Length Sync is off, Length moves to fit the file. If Sync is on, the length stays tempo-derived and only the newest part of the file is kept.
- **Unfreezing after a load replaces the loaded sample** with live input within one buffer length. Save anything you want to keep.
- A loaded sample is saved with your session automatically. Fractus stores the audio itself, not a link to the file, so moving or deleting the original file doesn't matter.

9.3 PRESETS

Control	What it does
Browse	Shows the current preset's name. Click it to choose a preset.
Save Preset	Saves the current settings as a new preset.
With Buffer	When ticked, the preset also stores the frozen buffer's audio. Available only while frozen.

See Chapter 11.

9.4 MIDI

MIDI Mode, the Amp Envelope (Attack, Decay, Sustain, Release), MIDI Legato, MIDI Velocity and **MIDI Reset**. See Chapter 10.

10. MIDI

Fractus works fully with no MIDI connected. To use MIDI, route a MIDI track to Fractus. In most DAWs you do this by choosing Fractus as the MIDI track's output or destination.

- **Nuendo / Cubase:** insert Fractus on the audio track, create a MIDI track, and set the MIDI track's **output routing** to Fractus.
- **Pro Tools:** insert Fractus on the audio track, create a MIDI (or Instrument) track, and set its **output** to the Fractus insert, listed by track name and insert.
- **Logic Pro:** Fractus is a **MIDI-controlled effect**. Insert it in an audio track's Audio FX slot, then in the Fractus window's header choose the software instrument track to take MIDI from, using the **Side Chain** menu. Play or record notes on that instrument track.
- **Ableton Live:** create a MIDI track, set **MIDI To** to the audio track that holds Fractus, then choose Fractus in the second menu below it.

The MIDI Learn menu (right-click any control) works as soon as MIDI reaches Fractus, in every MIDI Mode, including Off.

10.1 MIDI Mode

Mode	Behaviour
Off (default)	Notes are ignored. Fractus is an always-on effect. MIDI Learn still works.
Pitch	Notes retune the grain cloud. Nothing is gated, so the cloud drones on and keeps the pitch of the last note played.
Pitch + Gate	A monophonic, pitch- and velocity-tracking synth. The cloud sounds only while a note is held, shaped by the Amp Envelope.
Poly	Up to 16 notes at once, each retuning its own share of the grains. Not gated, so chords drone and hold their pitches.
Poly + Gate	A 16-voice polyphonic instrument. Each note has its own envelope.
Keyswitch	Notes select the sync divisions instead of playing (section 10.5).

- Notes **add to** the Pitch control, with middle C (MIDI note 60) as no transposition. Pitch then works as a fine-tune or offset.
- In the **Poly** modes, grain density rises with the number of held notes, so **a chord gets denser and richer, not louder**.
- **Gating comes before the diffuser and reverb**, so releasing a note silences the grains while the reverb tail rings on naturally.

10.2 Amp Envelope

Control	Range
Attack	0.1 ms to 5 s
Decay	0.1 ms to 5 s
Sustain	0 to 100%
Release	0.1 ms to 5 s

Control	Range
MIDI Velocity	How much note velocity affects level, from 0 to 100%.
MIDI Legato	In the mono modes, overlapping notes glide without restarting the envelope. Legato has no effect in the Poly modes and dims there.

- In the mono modes, releasing the newest note falls back to any note still held (last-note priority).
- Retriggering restarts the attack from the envelope's current level, so there is no click. At Sustain 100% a retrigger is inaudible.

10.3 Sustain pedal

CC64 holds notes in the gated modes, as on a piano. Releases played while the pedal is down wait until you lift it.

10.4 MIDI Reset (panic)

MIDI Reset on the ADVANCED page releases every note and the pedal immediately. Your host's own panic button (which sends All Notes Off) does the same.

10.5 Keyswitch mode

In Keyswitch mode, notes select divisions instead of playing. Note names use the **C3 = middle C (60)** convention:

Notes	Selects	Mapping
C1 to C#2 (MIDI 36–49)	Trigger Division	C1 = 1/1, C#1 = 1/2, ... in list order, getting shorter as notes rise
C3 to C#4 (MIDI 60–73)	Grain Size Division	C3 = 1/1, C#3 = 1/2, ...

- A division can only be keyswitched **while its sync is on**: Trigger Mode set to Sync for the Trigger range, Grain Size Sync on for the Grain range. The Keyswitch entry in the MIDI Mode list is greyed out until at least one of these is on.
- Changes land **exactly on the note**, to the sample.

- The selection **moves the on-screen control and records as automation** in your DAW's write modes. In Read mode, an automation lane on the same Division will fight the keyswitch.
- The switch latches: note-offs and velocity are ignored.
- The Amp Envelope, Velocity, Legato and MIDI Reset dim in Keyswitch mode.

10.6 MIDI Learn

Any MIDI controller knob, fader or pedal can take over any Fractus control.

1. **Right-click** the control and choose **MIDI Learn**. The control shows a violet *listening* ring and a banner at the top names it.
 2. **Move a knob** on your controller. It is now mapped.
 3. To cancel while listening, click the banner or right-click and choose **Stop Listening**.
 4. To remove a mapping, right-click and choose **Clear MIDI Learn (CC n)**.
 - Your controller covers the **control's full range**, from minimum to maximum.
 - Moving it **moves the on-screen control and records automation** like a mouse move.
 - **A learned control and an automation lane in Read mode will fight**, just as grabbing the knob with the mouse during playback does.
 - One CC can drive several controls. Learning a new CC for a control replaces the old one.
 - CC64 (sustain) and CC120–127 (channel mode messages) cannot be learned.
 - MIDI Learn works in every MIDI Mode, including Off, and listens on all channels.
 - Mappings are saved with your **session**, not with presets.
-

11. Presets, sessions and the frozen buffer

Saving with your session

Fractus saves everything with your DAW project. **If the buffer is frozen, its audio is saved too**, so reopening the project brings back the frozen material and its waveform. An unfrozen buffer is not saved: it is only a rolling window of whatever was playing.

A frozen buffer adds some size to your project: about 1.5 MB for 4 seconds at 48 kHz, up to about 44 MB for 30 seconds at 192 kHz.

Presets

- A preset stores **the patch**: every control setting.
- **With Buffer** (available while frozen) also stores the frozen audio, so a patch and its material travel as one file.

- A preset **never changes your setup**: loading one leaves your MIDI Learn mappings and theme as they are.
 - **Mind the size of With Buffer presets**. A 30-second buffer at 96 kHz is about 22 MB per preset. Use With Buffer only when the material is the point.
 - **Preset folder**:
 - macOS: `~/Library/Application Support/The WAV Lab/Fractus/Presets` (your own Library folder, which Finder hides: see Chapter 14)
 - Windows: `%APPDATA%\The WAV Lab\Fractus\Presets`
 - Preset files use the `.fractus` extension and can be copied between computers and platforms.
 - **Factory presets** arrive with version 1.0; this beta has none. They will never carry a buffer, since they are designed to work on your own audio.
-

12. Tips and techniques

- **Frozen pad**: play a chord, press Freeze, then set Jitter to 60–100%, Grain Size to 150–400 ms, Scatter to 70%, Grain Shape to 100% (Hann), and use the Plate reverb with a long Decay.
 - **Rhythmic stutter**: set Trigger Mode to Sync, Division to 1/16, Grain Size shorter than a 1/16 (for example 40 ms) and Grain Shape at 0%. Shorter grains make the chop more obvious.
 - **Reverse swell**: put Position on a transient, set Reverse to 100% and Grain Size to 300–600 ms.
 - **Octave shimmer**: set Pitch to +12, Jitter to 20%, Grain Mix to 60% and raise Reverb Mix.
 - **Moving texture**: route a slow LFO to Position and Random 1 (with some Slew) to Pitch at low depth.
 - **Tape memory**: set Character to 40% with Bit Crush at 0%, so you hear only saturation and band-limit.
 - **Play the buffer**: use Poly + Gate, freeze a sustained sound and play chords on it.
 - **Tempo-locked granular delay**: set Grain Size Sync to 1/8 and Trigger Sync to 1/8 at Position 0, then add Jitter slowly.
 - **Keep it tidy on mono sources**: Scatter and Pan build the stereo image from a mono track.
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13. Things that may surprise you

These are all by design.

1. **"Density doesn't do anything above a certain point."** Max Grains is limiting the rate (section 7.5). Density dims and shows the rate actually delivered. Raise Max Grains.
 2. **"My Sync division sounds slower than it should."** The same limit, in Sync mode: the grid is thinned to every 2nd or 3rd point and the Division dims. Raise Max Grains or shorten Grain Size.
 3. **"I can't hear the rhythm in Sync mode."** Your grains are longer than the Division, so they overlap into a continuous sound. Shorten Grain Size below the Division to hear the grid.
 4. **"Fractus doesn't sound granular at all."** With Position 0, Jitter 0 and no pitch shift, every grain reads the same live audio and the result is close to the dry signal. Move Position, raise Jitter or shift Pitch.
 5. **"Grains end early in the display."** Grains have to fit inside the recorded audio. At high pitch, when reversed, or while frozen, long grains need more room, so they are shortened with a fade. Raise Buffer Length for long, pitched or reversed grains. Reverse grains need about twice the room of forward grains.
 6. **"Position won't reach the ends when pitched or frozen."** Same reason: the extremes need room for the grain to play. For example, the reach shrinks by about 2% of a default buffer at +1 semitone, and 15% at +12 semitones with 500 ms grains.
 7. **"Very low Density is quieter."** Once grains are sparse enough not to overlap (below about 12 Hz with 80 ms grains), each plays at full level with silence between. Nothing is boosted above that, to avoid clipping.
 8. **"Fractus went quiet after I changed my audio interface's sample rate."** Changing sample rate empties the buffer. If Freeze was on, nothing can refill it. Turn Freeze off, then refreeze. Changing the **buffer size** is safe.
 9. **"The Character stage controls don't do anything."** Character (the macro) is at 0%.
 10. **"The loaded sample disappeared."** Freeze was turned off, and live input replaced it within one buffer length.
 11. **"My session reopened with Freeze off."** If a saved session carried no buffer audio, for example because the audio couldn't be read, Freeze is turned off automatically so the plugin isn't stuck silent.
 12. **"Freezing clicked once."** Long grains already playing at the moment you freeze can click once each, for up to one grain length.
 13. **"A mono source sounds a little hotter in the reverb."** The Plate and Early Reflections algorithms take a mono sum of their input, as classic plate hardware does. Mono or dual-mono material lands up to about 3 dB hotter in their tail at full Reverb Mix (about 0.6 dB at the defaults). The FDN is unaffected.
 14. **"A lowered Max Grains still shows extra grains."** Grains already playing are allowed to finish and fade out rather than being cut off.
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14. Troubleshooting

Problem	Try
Fractus doesn't appear in my DAW	Restart the DAW after installing. Rescan plug-ins. Check that the format your DAW needs is installed. On a Mac, Fractus requires Apple Silicon and macOS 12 or later.
An update didn't take effect	Fully quit and restart the DAW.
Silent after inserting	Is Freeze on with an empty buffer? Turn Freeze off. Is MIDI Mode set to a Gate mode with no notes playing? Set it to Off.
Regular short dropouts	Demo mode: no trial has been started, or the trial has ended. Start the trial or unlock with your licence key (Chapter 3).
The unlock page says my key is on two computers already	Use Reset my computers on the unlock page (once a month), then unlock this computer.
My response code stopped working after an OS reinstall	Reinstalling can change the machine code. Reset your computers on the unlock page and unlock again.
Silent after a sample-rate change	Toggle Freeze off (see Chapter 13, item 8).
Stuck notes	Press MIDI Reset on the ADVANCED page, or use your host's panic button.
High CPU	Lower Max Grains. Run at 44.1 or 48 kHz rather than 96 or 192 kHz. Keep Density × Grain Size moderate.
A MIDI controller knob doesn't respond	Is it CC64 or 120–127? Those can't be learned. Is MIDI routed to Fractus?
Automation and a MIDI-learned knob fight	Set the automation lane to Touch, Latch or Off while performing.
I can't find the preset or licence folder on a Mac	It is in your own Library folder, which Finder hides, not the Library folder at the top of the disk. In Finder choose Go → Go to Folder...

Problem	Try
	(Shift-Command-G), paste ~/Library/Application Support/The WAV Lab/Fractus and press Return. The licence is the file <code>license.lic</code> ; presets are in <code>Presets</code> .
Pro Tools: Fractus is missing from a track	Check that the track is mono or stereo, and that the AAX version is installed.

Support: email info@thewavlab.com. Please include your operating system, DAW and version, the Fractus version (shown in the About box), and what you were doing when the problem happened.

15. Credits and acknowledgements

Fractus is designed and built by **The WAV Lab** (Sounds Like Joe LLC).

Fractus is an independent, original design. It is **inspired by Mutable Instruments and Émilie Gillet**, whose Clouds module showed how musical a granular processor can be, and by the pioneers of granular synthesis: **Dennis Gabor, Iannis Xenakis, Curtis Roads and Barry Truax**. No Mutable Instruments code is used. All of Fractus's signal processing, including its reverbs, was written from scratch.

The interface uses **IBM Plex Sans and IBM Plex Mono**, licensed under the SIL Open Font License 1.1. Licence signatures are checked with **Monocypher** (BSD 2-clause).

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